INCH-POUND

MIL-DTL-55302/117E
9 January 2004
SUPERSEDING
MIL-C-55302/117D
6 May 1993

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

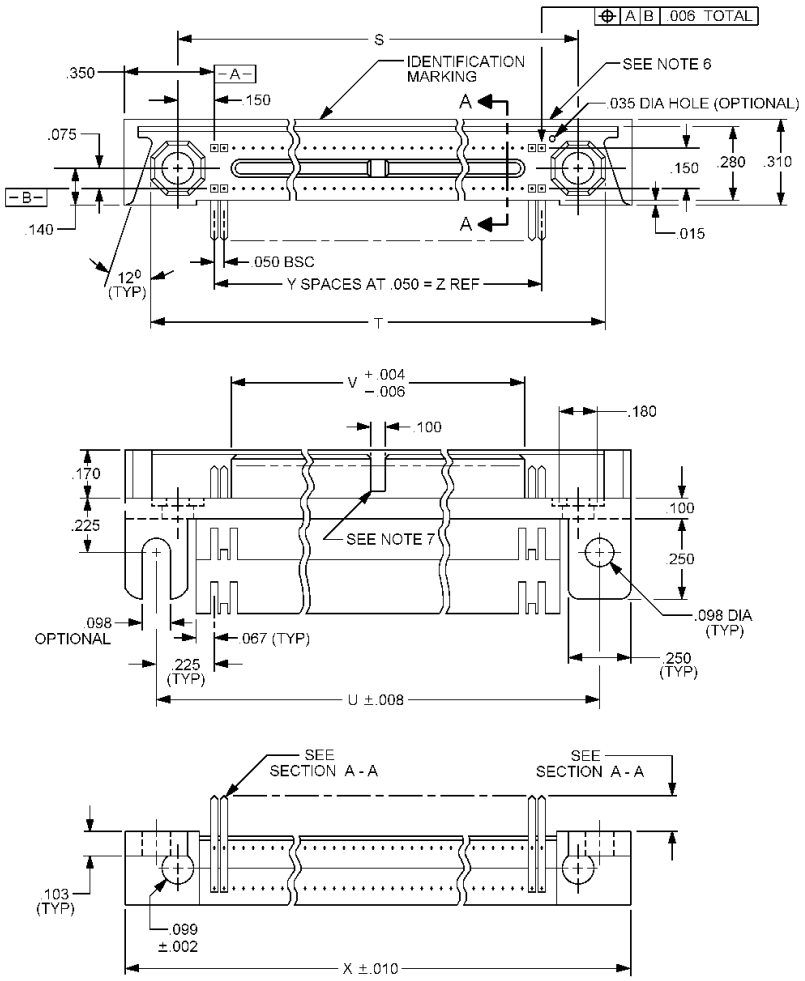

FIGURE 1. Connectors, plug .050 (1.27 mm) square grid.

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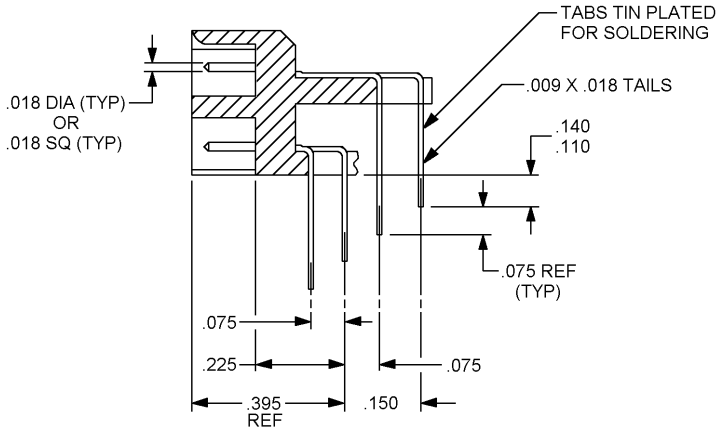
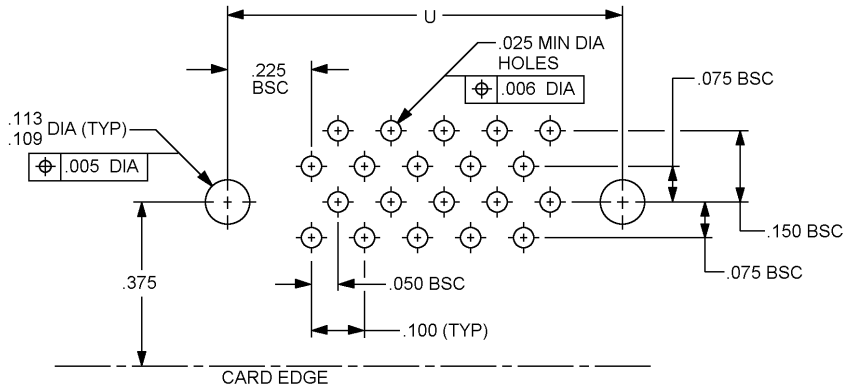


SECTION A - A



RECOMMENDED PRINTED WIRING BOARD LAYOUT

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.035	0.89	.140	3.56
.004	0.10	.050	1.27	.150	3.81
.005	0.13	.067	1.70	.180	4.57
.006	0.15	.075	1.90	.225	5.72
.008	0.20	.098	2.49	.250	6.35
.009	0.23	.100	2.54	.280	7.11
.010	0.25	.103	2.62	.310	7.87
.015	0.38	.109	2.77	.350	8.89
.018	0.46	.110	2.79	.375	9.52
.025	0.64	.113	2.87	.395	10.03

FIGURE 1. Connectors, plug .050 (1.27 mm) square grid - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-DTL-55302/119.
5. Pads suitable for printed wiring board support are optional.
6. Numbers indicating end positions stamped this side only.
7. Void is not present on 20 through 50 positions.

FIGURE 1. Connectors, plug .050 (1.27 mm) square grid - Continued.

TABLE I. Dash numbers and dimensions.

Dash number	Number of contacts	Dimensions							Contact Identification numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.50)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
04	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
05	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
07	80	2.250 (57.15)	2.470 (62.47)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
09	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
10	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
11	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Material:

Insulator body: Insulator material shall be in accordance with MIL-DTL-55302 or ASTM D5138 type LCP0120G30A43430 EA300ED035EE200EF140 or MIL-M-24519 type GLCP-30F.

Contact material: Contact material shall be in accordance with MIL-DTL-55302 or beryllium copper alloy C17400 in accordance with ASTM-B768.

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Plating: Contact plating shall be gold over nickel in accordance with MIL-DTL-55302 on the contact engagement area for .100 inch (2.54 mm) minimum length. The termination area plating shall be tin lead over nickel in accordance with MIL-DTL-55302 to within .020 inches maximum of plastic housing body. The remaining areas of the contact shall be nickel plated in accordance with MIL-DTL-55302.

Contact identification: See figure 1.

Contact rating: 1.5 Amperes per contact, maximum.

Operating temperature: -65°C to +125°C.

Keying: See MIL-DTL-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have resistance exceeding .030 ohm.

Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only) (applicable to qualification and retention of qualification only).

Part or Identifying Number (PIN): M55302/117 - (dash number from table 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government of for use in equipment to be delivered to the Government.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/31	MIL-STD-202
MIL-DTL-55302/119	ASTM B768
MIL-M-24519	ASTM D5138

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4469-000)

Review activities:
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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